

Sayantan Pal

Doctoral Student

Unmanned Vehicle System Lab, Department of Aerospace Engineering, Indian Institute of Technology Kharagpur

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Professional Summary

Researcher with expertise in guidance, and control of autonomous aerial vehicles, focusing on nonlinear control, vector field guidance, path planning, and obstacle avoidance. Experienced in developing robust guidance algorithms for UAVs and missile systems, with publications in leading IEEE journals and international conferences. Research interests include autonomous systems, intelligent guidance, optimal control, and aerospace autonomy.

Research Experience

Indian Institute of Technology Kharagpur, Kharagpur, India

Aug 2021 – Present

Graduate Research Assistant, Department of Aerospace Engineering

- Conducting doctoral research on guidance, and control (GNC) of autonomous aerial vehicles.
- Developing nonlinear guidance algorithms for UAV path following, obstacle avoidance, curvature-constrained trajectory generation, and impact-angle-constrained missile guidance.
- Designing vector field guidance laws and Lyapunov-based control strategies with theoretical stability guarantees under actuator and field-of-view constraints.
- Validating guidance and control algorithms through MATLAB/Simulink simulations and experimental implementation on Quanser QDrone and OptiTrack motion capture systems.

CSIR–Central Mechanical Engineering Research Institute (ATDC), Durgapur, India

Dec 2020 – Jul 2021

Project Associate

- Contributed to the design and development of a tendon-driven robotic hand exoskeleton for post-stroke rehabilitation.
- Developed mechanical components and evaluated prototype performance for assistive rehabilitation applications.

CSIR–Central Mechanical Engineering Research Institute, Durgapur, India

Jul 2019 – Aug 2020

Research Assistant

- Conducted research on flexible and stretchable strain sensors for wearable robotic exosuits intended for lower-limb assistance.
- Designed, fabricated, and experimentally characterized conductive fabric-based soft sensors and pneumatic artificial muscles.
- Contributed to multiple peer-reviewed publications in wearable robotics and soft sensing technologies.

Core Competencies

Control & Guidance	Flight Control Systems, Missile Guidance, Waypoint Following, UAV Formations, Obstacle Avoidance
Programming	MATLAB/Simulink, Python, C/C++, L ^A T _E X, Git
Experimental Platforms	Quanser QDrone, Quanser QBot, OptiTrack Motive
CAD & Design	CATIA, SolidWorks
Machine Learning	TensorFlow, Machine Learning Fundamentals
Languages	English (Professional), Bengali (Native), Hindi (Professional)

Education

Indian Institute of Technology Kharagpur (IIT Kharagpur)

2021–Present

Ph.D., Aerospace Engineering

Indian Institute of Engineering Science and Technology (IIST), Shibpur

2018–2020

M.Tech., Mechatronics

Maulana Abul Kalam Azad University of Technology (MAKAUT), Kolkata

2012–2016

B.Tech., Mechanical Engineering

Selected Publications

Journal Articles

- **S. Pal** and S. Hota, “Curvature-Constrained Vector Field Guidance for UAVs,” *IEEE Transactions on Aerospace and Electronic Systems*, 2026.
- **S. Pal** and S. Hota, “Impact Angle Control Guidance Using Asymmetric Integral Barrier Lyapunov Function with FOV and Input Constraints,” *IEEE Control Systems Letters*, vol. 9, pp. 2663–2668, 2025.
- P. Mahajan, **S. Pal**, and S. Hota, “Continuous-Curvature Dubins-Inspired L1 Guidance with Sliding Mode Control-Based Tracking for UAVs,” *Proceedings of the Institution of Mechanical Engineers, Part G: Journal of Aerospace Engineering*, 2025.

Conference Proceedings

- **S. Pal** and S. Hota, “Arccosine Vector Field Guidance for Curvilinear Path Following of UAVs,” *International Conference on Control, Decision and Information Technologies (CoDIT)*, 2026. (Accepted)
- I. Phanse, **S. Pal**, and S. Hota, “Varying-Gain Lyapunov Guidance Vector Field for Elliptical Path Following,” *IEEE SPACE Conference*, 2026. (Accepted)
- P. Madure, G. Khurd, **S. Pal**, and S. Hota, “Guidance Law for Surveillance by UAVs,” *IEEE SPACE Conference*, 2026. (Accepted)
- K. S. K. Hrithika, **S. Pal**, and S. Hota, “UAV Path Following with Obstacle Avoidance Using Vector Field Approach,” *INSTCon*, 2026. (Accepted)
- P. P. Kumar, **S. Pal**, and S. Hota, “Barrier Sliding Mode Control for Curvature-Constrained Nonlinear Planar Path Following of UAVs,” *IEEE Conference on Decision and Control (CDC)*, 2026. (Accepted)
- P. Mahajan, **S. Pal**, and S. Hota, “Time-Optimal L1 Guidance for Straight-Line Path Following in 3D,” *International Conference on Mechatronics and Robotics Engineering (ICMRE)*, 2026.
- N. K. Singh, **S. Pal**, A. D. Menon, and S. Hota, “Impact Angle Constrained Guidance Laws for Stationary Target Interception in 3D Space,” *AIAA SciTech Forum*, 2026.
- **S. Pal** and S. Hota, “Vector Field Guidance with Curvature Constraint for UAVs,” *AIAA SciTech Forum*, 2025.
- N. K. Sahu, **S. Pal**, and S. Hota, “Impact Angle Constrained 3D Guidance Law for Stationary Target Interception,” *IEEE SPACE Conference*, 2025.
- S. Singha, **S. Pal**, and S. Hota, “Smooth Trajectory Generation for Gap Traversal by UAVs,” *IEEE SPACE Conference*, 2025.
- P. P. Kumar, **S. Pal**, and S. Hota, “Path Convergence of UAVs via Fast Terminal Sliding Mode Control,” *11th Indian Control Conference (ICC)*, 2025.
- **S. Pal** and S. Hota, “Nonlinear Control Law for Path Following by UAVs,” *10th Indian Control Conference (ICC)*, pp. 476–481, 2024.
- **S. Pal** and S. Hota, “Time-Optimal Bézier Curves for Straight-Line and Circular Path Convergence in 3D Space,” *9th Indian Control Conference (ICC)*, pp. 365–370, 2023.
- **S. Pal**, D. Sarkar, S. S. Roy, A. Kumar, and A. Arora, “Development of a Stretchable and Flexible Conductive Fabric-Based Sensorized Pneumatic Artificial Muscle,” *4th International Conference on Electronics, Communication and Aerospace Technology (ICECA)*, 2020.
- **S. Pal**, D. Sarkar, S. S. Roy, A. Paul, and A. Arora, “Design, Development and Analysis of a Conductive Fabric-Based Flexible and Stretchable Strain Sensor,” *IOP Conference Series: Materials Science and Engineering*, 2020.
- D. Sarkar, **S. Pal**, S. S. Roy, A. Kumar, and A. Arora, “Estimation of Transmission Force in Assistive Devices Using Conductive Liquid Metal-Based Sensorized Pneumatic Artificial Muscle,” *IEEE International Conference on Recent Advances and Innovations in Engineering (ICRAIE)*, 2020.
- A. Arora, D. Sarkar, **S. Pal**, A. Kumar, S. Sen, and S. S. Roy, “A Multi-Technique Measurement and Estimation of Volume for Pneumatic Artificial Muscle,” *IEEE International Conference on Recent Advances and Innovations in Engineering (ICRAIE)*, 2020.

Awards & Professional Activities

Awards & Honors

- **IEEE American Control Conference (ACC) Travel Grant**, 2026 – Awarded competitive travel and lodging support to present research at the IEEE American Control Conference (ACC), USA.
- **IIT Kharagpur International Travel Grant**, 2026 – Supported paper presentation at the IEEE International Conference on Control, Decision and Information Technologies (CoDIT), Bari, Italy.
- **IIT Kharagpur International Travel Grant**, 2025 – Supported paper presentation at the AIAA SciTech Forum,

Orlando, Florida, USA.

- **IEEE Indian Control Conference (ICC) Travel Grant**, 2023 – Awarded competitive travel funding for paper presentation at the 9th Indian Control Conference.
- **GATE Fellowship**, Government of India (2018–2021).

Professional Activities

- Reviewer for *IEEE Control Systems Letters*, *International Journal of Control, Automation and Systems*, IEEE Indian Control Conference (ICC 2025), and AIAA SciTech 2025.
- Graduate Student Member, IEEE.
- Volunteer, *Advances in Robotics and AI 2020* (CSIR-CMERI) and *ICTACEM 2026* (IIT Kharagpur).